

LC 1510: Stone Game IV

MICROSOFT

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| The Game & Rules

- Basically Aadya starts first.
- Both I and her play optimally.
- **Rule:** Can remove any non-zero square number from the pile, the first one to reach the N^{th} stone wins the game.
- In other words, the player who can't make any more moves loses the game.

| Dynamic Programming Logic

- Let $dp[i]$ store whether the current player can win when i stones remain.
- Since we don't need to keep a track of both players' scores, we don't need $dp[i][0 \text{ or } 1]$.
- Now say x^2 stones are removed.
- Then $dp[i - x^2] \rightarrow$ for the other player.
- Now if the other player loses, the current player wins, so $dp[i] = \text{true}$.
- Hence, $dp[i] = \text{true}$ if for any x we can show $dp[i - x^2] = \text{false}$.
- So **Bottom-Up DP** with a base case of $dp[0] = \text{false} \rightarrow$ no stone left, can't win.

| Code Implementation

```
class Solution {
public:
    bool winnerSquareGame(int n) {

        //should be marked ezzzz
        //just use dp[i] to see if player can win when i stone left
        //say x^2 stone remove then next player will hve i-x^2 stones
        //if next player cant win currplayer wins
        //since for dp[i] we check dp[i-x^2] bottom up
        //base: dp[0]=false; 0 stones left cant win

        vector<int> dp(n+1,false); //initially koi nhi jeet skta
                                   //koi tabhi jeet skta agar next player haar
        dp[0]=false;
        for(int i=1;i<=n;i++){
            for(int x=1;x*x<=i;x++){

                if(dp[i-x*x]==false){
                    dp[i]=true;
                }
            }
        }
        return dp[n];
    }
};
```